Closed Loop Control of Single-phase Boost PFC Converter - II

Introduction

In today's energy landscape, efficient and reliable power conversion is paramount, especially with the proliferation of electric vehicle charging infrastructure and renewable energy integration. A critical component in achieving this is the single-phase boost Power Factor Corrector (PFC) converter. This lecture delves into the intricate details of its closed-loop control, building upon previous discussions. The primary objectives for operating such a converter are twofold: first, to maintain a precisely regulated output voltage, ensuring stable power delivery despite disturbances or variations in plant parameters; and second, to draw current from the AC grid with a unity power factor, minimizing reactive power and improving overall system efficiency. Achieving these goals requires a sophisticated control strategy, which necessitates a deep understanding of the converter's dynamic behavior. This session focuses on developing the foundational mathematical models, specifically the average large signal model using state-space representation, which are indispensable for designing robust and effective controllers.

Core Concepts

Term	Definition	Significance
`Gi(s)`	A transfer function representing how the inductor current in the boost PFC converter responds to changes in the duty ratio of its switches.	Crucial for designing the inner current control loop, ensuring the actual current tracks the reference.
`Gv(s)`	A transfer function describing how the output voltage of the converter varies in response to changes in the inductor current.	Essential for designing the outer voltage control loop, maintaining a stable and regulated output voltage.
`State Variables`	A minimal set of variables that completely describe the dynamic state of a system at any given time. For the boost	Form the basis for state-space modeling, allowing for a comprehensive mathematical description

Term	Definition	Significance
	PFC, these are inductor current (`iL`) and capacitor voltage (`vc`).	of the converter's behavior.
`Average Large Signal Model`	A simplified mathematical model that represents the average behavior of a switching converter over a switching period, ignoring high-frequency ripple.	Provides a tractable model for control design by abstracting away fast switching dynamics, focusing on slower, average responses.
`State-Space Representation`	A mathematical model of a physical system as a set of input, output, and state variables related by first-order differential equations.	A powerful tool for analyzing and designing controllers for complex dynamic systems like power converters.
`DTs Period`	The duration within a switching cycle (`Ts`) during which the main switch (S1) of the boost converter is ON, determined by the duty ratio `D` .	Defines one of the two distinct operating modes of the converter, each with its own set of state equations.
`(1-D)Ts Period`	The remaining duration within a switching cycle (`Ts`) during which the main switch (S1) is OFF and the diode (S2) conducts.	Defines the second distinct operating mode, completing the full switching cycle and contributing to the overall average behavior.
`Linearization`	The process of approximating a non-linear system with a linear model around a specific operating point.	Simplifies complex non-linear converter dynamics into a form suitable for conventional linear control theory design.

Detailed Analysis

1. Recap of Closed-Loop Control Objectives and Transfer Functions

The overarching goal of closed-loop control for the single-phase boost PFC converter is to achieve both a regulated output voltage and unity power factor operation. To design effective controllers, we first need to understand the converter's dynamic response. This is typically broken down into two key transfer functions:

- **`Gi(s)` (Current Loop Transfer Function):** This function quantifies how the inductor current (`iL`) responds to changes in the duty ratio (`D`) of the converter's switches. It is fundamental for designing the inner current control loop, which ensures that the actual inductor current accurately tracks its reference.
- **`Gv(s)` (Voltage Loop Transfer Function):** This function describes how the output voltage (`Vo` , which is equivalent to the capacitor voltage `vc`) changes in response to variations in the inductor current. It is crucial for designing the outer voltage control loop, responsible for maintaining the output voltage at the desired regulated value.

By defining these transfer functions, we can systematically determine the parameters for current and voltage controllers, ensuring objectives like required phase margin, gain margin, and disturbance rejection ratios are met. The current controller ensures the actual current follows the set reference, while the voltage controller works to minimize the error between the desired and actual output voltage.

2. Deriving the Average Large Signal Model

To understand the converter's dynamics, we employ a systematic method to derive its transfer functions. This involves:
* **Average Large Signal Model:** First, we develop an average large signal model using state equations. This model averages the converter's behavior over a switching period, simplifying the analysis by removing high-frequency switching ripple.
* **Small Perturbation:** Next, we introduce small perturbations around a steady-state operating point. This allows us to analyze the system's response to small changes.
* **Linearization:** We then linearize the state equations around this operating point. Since power converters are inherently non-linear, linearization simplifies the model, making it amenable to linear control theory.
* **Time-Domain to S-Domain Conversion:** Finally, the linearized time-domain

differential equations are converted into the s-domain (Laplace domain) to obtain the desired transfer functions, `Gi(s)` and `Gv(s)`.

3. State Variables and Input/Output Definition

For the single-phase boost PFC converter, there are two energy storage elements: the inductor (L) and the capacitor (C). These elements define the system's state variables:
* **Inductor Current (`iL`):** The current flowing through the inductor, which is also the input current (`id`) to the boost stage. Inductors resist sudden changes in current.
* **Capacitor Voltage (`vc`):** The voltage across the output capacitor, which is also the output voltage (`Vo`). Capacitors resist sudden changes in voltage.

The inputs to the system that influence these state variables are:
* **Duty Ratio (`D`):** The control input, which determines the ON-time of the main switch.
* **Input Voltage (`vs` or `vd`):** The rectified input AC voltage, `|vs(t)|`.

The primary output of interest is the **Output Voltage (`Vo`)**, which is equal to `vc`.

4. State-Space Representation during DTs Period

During the `DTs` period (when switch S1 is ON and S2 is OFF):
* The inductor is connected directly across the rectified input voltage `|vs|`.
* The load current is supplied entirely by the output capacitor.

The state equations are derived as follows:
* **Inductor Voltage:** `vL = L * diL/dt = |vs|`.
* Therefore, `diL/dt = |vs| / L`.
* **Capacitor Current:** The capacitor current `ic` is the negative of the load current. The load current is `Vo / RL`, and since `Vo = vc`, `ic = -vc / RL`.
* Also, `ic = C * dVc/dt`.
* Therefore, `dVc/dt = -vc / (RL * C)`.

In state-space form, with `x = [iL; vc]` and input `u = |vs|`:
`x_dot = A1 * x + B1 * u`
`Vo = C1 * x + D1 * u`

Where:
`A1 = [[0, 0], [0, -1/(RL*C)]]`
`B1 = [[1/L], [0]]`
`C1 = [0, 1]` (since `Vo = vc`)
`D1 = [0]`

5. State-Space Representation during (1-D)Ts Period

During the `(1-D)Ts` period (when switch S1 is OFF and S2 is ON):
* The inductor is connected across the difference between the input voltage `|vs|` and the output capacitor voltage `vc`.
* The inductor current `iL` flows to both the load `RL` and the capacitor `C`.

The state equations are derived as follows:
* **Inductor Voltage:** `vL = L * diL/dt = |vs| - vc`.
* Therefore, `diL/dt = (|vs| - vc) / L`.
* **Capacitor Current:** The capacitor current `ic` is the inductor current `iL` minus the load current `Vo / RL`. Since `Vo = vc`, `ic = iL - vc / RL`.
* Also, `ic = C * dVc/dt`.
* Therefore, `dVc/dt = (iL - vc / RL) / C = (1/C) * iL - (1/(RL*C)) * vc`.

In state-space form, with `x = [iL; vc]` and input `u = |vs|`:
`x_dot = A2 * x + B2 * u`
`Vo = C2 * x + D2 * u`

Where:
`A2 = [[0, -1/L], [1/C, -1/(RL*C)]]`
`B2 = [[1/L], [0]]`
`C2 = [0, 1]` (since `Vo = vc`)
`D2 = [0]`

6. Averaging the State-Space Models

To obtain the average large signal model, we combine the state-space representations from the `DTs` and `(1-D)Ts` periods, weighted by their respective durations:

`x_dot_avg = D * (A1 * x + B1 * |vs|) + (1 - D) * (A2 * x + B2 * |vs|)`
`Vo_avg = D * (C1 * x + D1 * |vs|) + (1 - D) * (C2 * x + D2 * |vs|)`

Rearranging these equations:
`x_dot_avg = (D * A1 + (1 - D) * A2) * x + (D * B1 + (1 - D) * B2) * |vs|`
`Vo_avg = (D * C1 + (1 - D) * C2) * x + (D * D1 + (1 - D) * D2) * |vs|`

Let `A = D * A1 + (1 - D) * A2`, `B = D * B1 + (1 - D) * B2`, and `C = D * C1 + (1 - D) * C2`.
Since `D1 = 0` and `D2 = 0`, the input term in the output equation becomes zero.

7. Final Average State-Space Model

Substituting the matrices A1, B1, C1, D1, A2, B2, C2, D2 into the averaged equations, we get the final average large signal state-space representation:

A Matrix:

`A = [[0, -(1-D)/L], [(1-D)/C, -1/(RL*C)]]`

B Matrix:

`B = [[1/L], [0]]`

C Matrix:

`C = [0, 1]`

Thus, the average large signal model is:

`[diL/dt]`
`[dVc/dt]` = `A` * `[iL]`
`[vc]` + `B` * `|vs|`

And the output equation is:

`Vo = C` * `[iL]`
`[vc]`

This derived model forms the basis for further analysis, including perturbation and linearization, to obtain the small-signal transfer functions necessary for controller design.

Key Takeaways

- [ ] The closed-loop control of a single-phase boost PFC converter aims for regulated output voltage and unity power factor.
- [ ] `Gi(s)` describes inductor current response to duty ratio changes, crucial for the inner current loop.
- [ ] `Gv(s)` describes output voltage response to inductor current changes, essential for the outer voltage loop.
- [ ] The two state variables for the boost PFC converter are inductor current (`iL`) and capacitor voltage (`vc`).
- [ ] The average large signal model simplifies converter dynamics by averaging over a switching period.
- [ ] State-space representation provides a structured mathematical framework for analyzing converter behavior.
- [ ] The converter operates in two distinct modes: `DTs` (S1 ON, S2 OFF) and `(1-D)Ts` (S1 OFF, S2 ON), each with unique state equations.
- [ ] The overall average state-space model is obtained by weighting and combining the state equations from these two operating modes.
- [ ] Linearization of the average model around an operating point is necessary to apply linear control theory.

- [ ] The derived A, B, and C matrices define the average large signal state-space model, which is the starting point for controller design.

Conclusion

The detailed derivation of the average large signal model using state-space representation is a foundational step in designing effective closed-loop controllers for the single-phase boost PFC converter. By accurately capturing the dynamic relationship between the converter's inputs (duty ratio, input voltage) and its state variables (inductor current, capacitor voltage), we create a mathematical framework that allows for rigorous analysis. This model, once linearized, will enable the calculation of critical transfer functions like `Gi(s)` and `Gv(s)`. These transfer functions are the blueprints for designing the inner current and outer voltage control loops, which are essential for achieving the converter's primary objectives: maintaining a stable output voltage and ensuring a high power factor. This systematic approach is vital for developing robust and efficient power electronics solutions, particularly in demanding applications such as electric vehicle charging infrastructure, where precise control and grid compatibility are paramount. The next steps will involve perturbing and linearizing this model to extract the small-signal transfer functions, ultimately leading to the design of the actual controllers.